KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
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Project Work Review - 2

Centre for Distance and Online Education

July -2024

IV Semester

MCA

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Project Title : MaatriCare

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Register number : 9224116032

Name of the Programme : MCA

Semester : IV

Name of the Course : Project Work

Batch : July 2024

Course Code : 223MCA4201

Regulation : 2022

ABSTRACT

MaatriCare is a pregnancy support and care-coordination web application that organises pregnancy profiles, appointments, daily care, wellbeing records, symptom journals, milestones, learning resources, reports, and reminders in one authenticated system. Users can prepare questions for healthcare visits, create a printable visit brief, and choose which records to include in a report.

The frontend is implemented with React, TypeScript, and Vite. The backend uses Java Spring Boot REST APIs, Spring Security with JWT authentication, request validation, and Spring Data JPA. MySQL supports local development, with PostgreSQL configured for production. Relational data includes user profiles, appointments, care tasks, wellbeing entries, symptoms, milestones, emergency contacts, report shares, push subscriptions, and scheduled reminders.

MaatriCare is an organizational and educational aid. It does not diagnose conditions, recommend treatment, or replace a healthcare professional or emergency services.

Introduction

1.Project Overview

MaatriCare is a private workspace for pregnancy-related organization and care coordination.

2.Objectives

1.- Keep pregnancy profile and progress information together.

2.Track daily care and wellbeing entries.

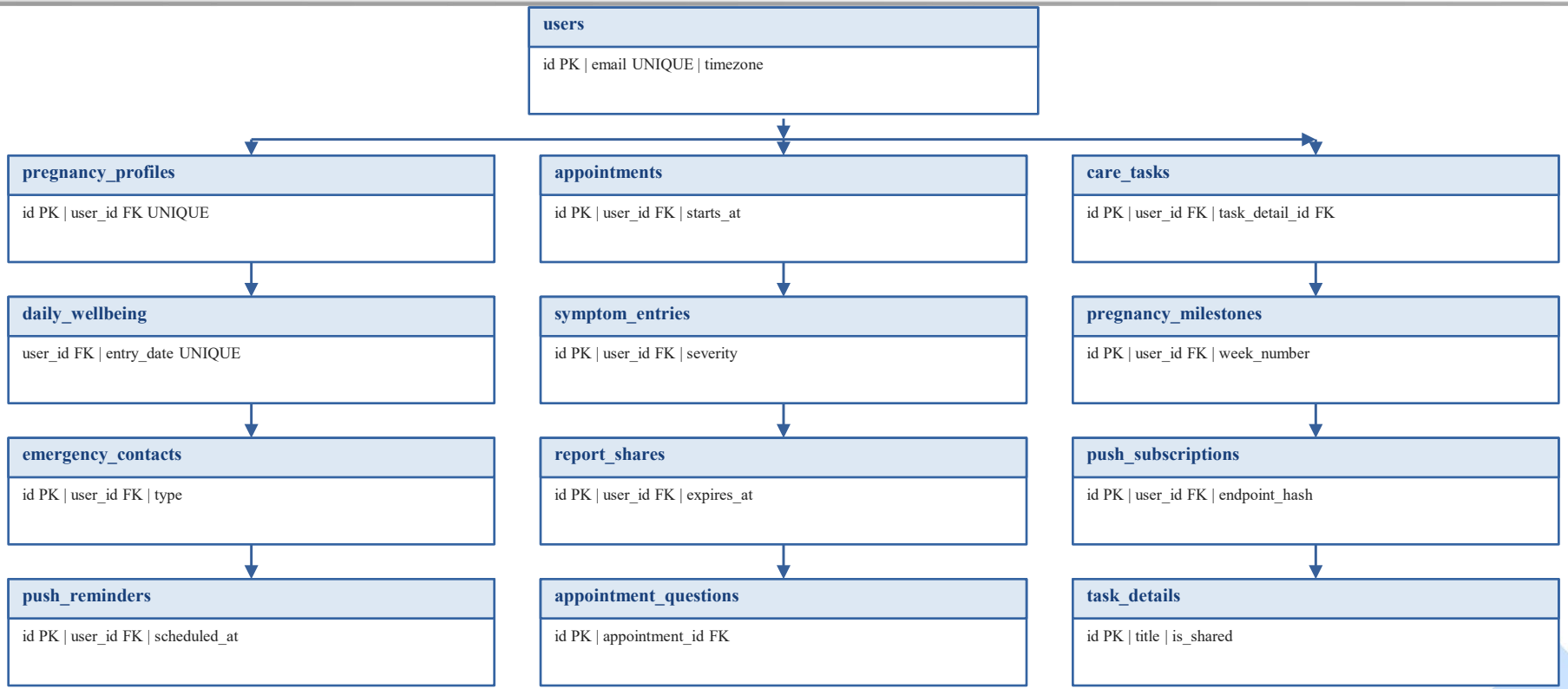
3.Manage appointments, questions, and visit briefs.

4.Provide educational resources, summaries, and configurable reminders.

3.Scope and Safety

The system supports user-entered records and healthcare conversations. It does not diagnose, prescribe, or provide emergency triage.

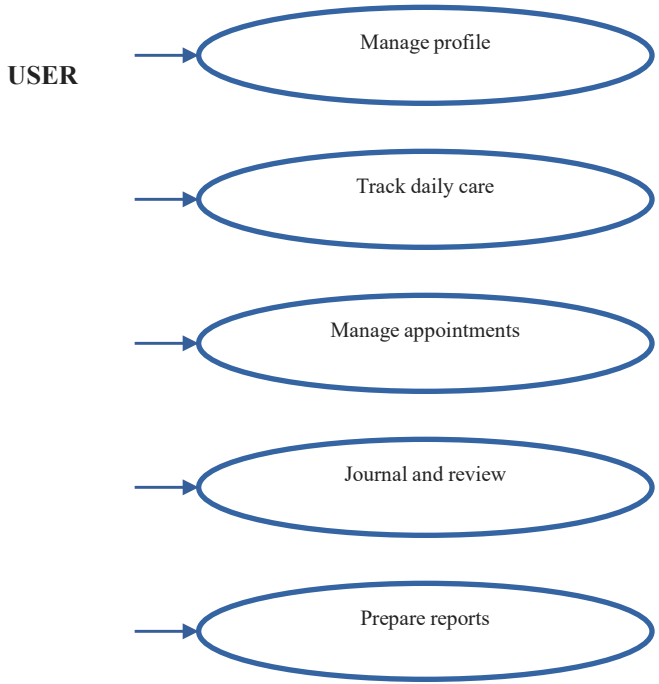
ER (Entity Relationship) Diagram



Relationships: users 1:N to user-owned records; pregnancy_profiles 1:1. appointment_questions reference appointments; care_tasks reference task_details.

UML Use-Case and Class Diagrams

USE CASES



CORE CLASSES

User

id | email | timezone

Appointment

user_id | starts_at | reminder

CareTask

user_id | date | completed

SymptomEntry

user_id | symptom | severity

PushReminder

user_id | scheduled_at | status

Hardware and Software Requirements

HARDWARE (DEVELOPMENT RECOMMENDATION)

PROCESSOR: Dual-core or better

RAM: 8 GB minimum; 16 GB recommended

STORAGE: 10 GB available

DISPLAY: 1366 x 768 or greater

NETWORK: Internet for package installation and backend testing

SOFTWARE REQUIREMENTS

OPERATING SYSTEM: Windows 10/11 or Linux

FRONT END: React, TypeScript, Vite

BACK END: Java 25, Spring Boot 3.5.5, Maven

DATABASE: MySQL local; PostgreSQL configuration; H2 tests

TOOLS: Node.js/npm, Git, modern browser

OPTIONAL: Ollama; VAPID configuration for Web Push

Module Name

Implemented modules include:

1) Authentication and pregnancy profile

2) Pregnancy progress and milestones

3) Daily care and Gentle Day Planner

4) Appointment and visit-question management

5) Private symptom journal

6) Guides and resource library

7) Reports and secure sharing

8) Emergency contacts

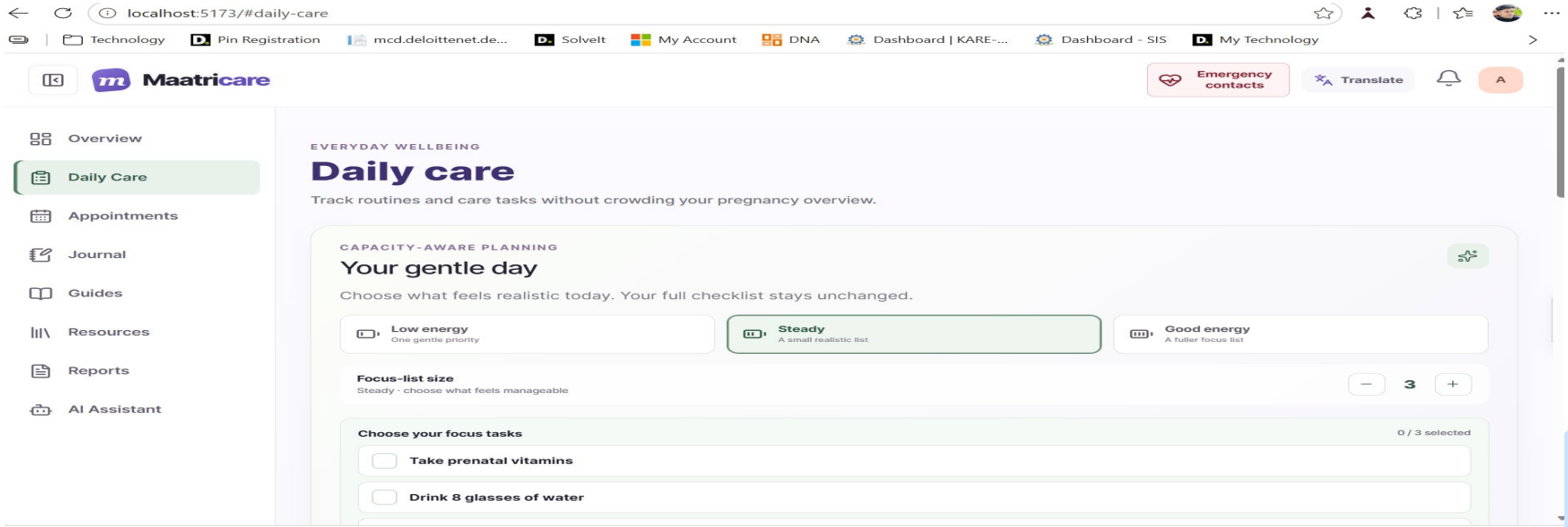
9) Reminder preferences and Web Push

Daily Care and Gentle Day Planner

The daily care module tracks user-selected routines, including hydration, vitamins, activity, sleep, and mood.

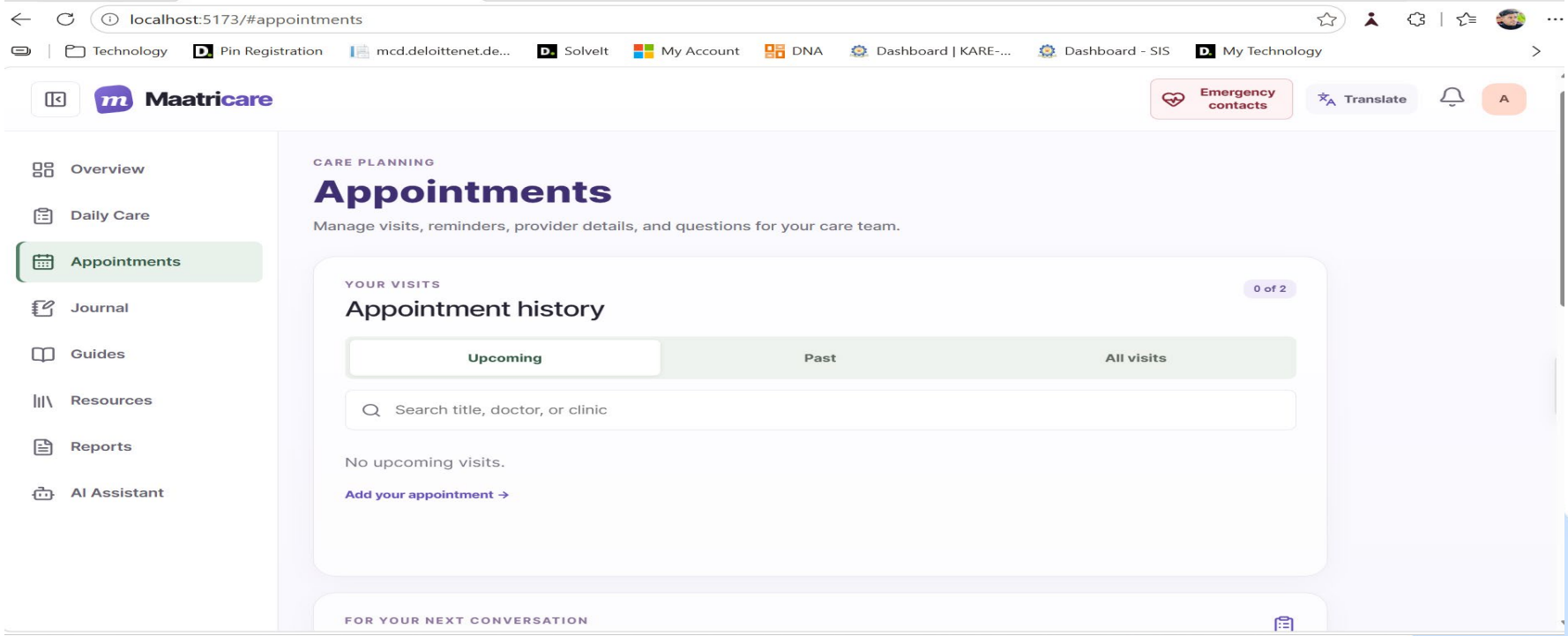
The Gentle Day Planner lets a user choose an energy level and a manageable focus list. Completion updates are saved to the account.

This module supports organization only; it does not prescribe a care plan.



Appointment and Visit Brief Module

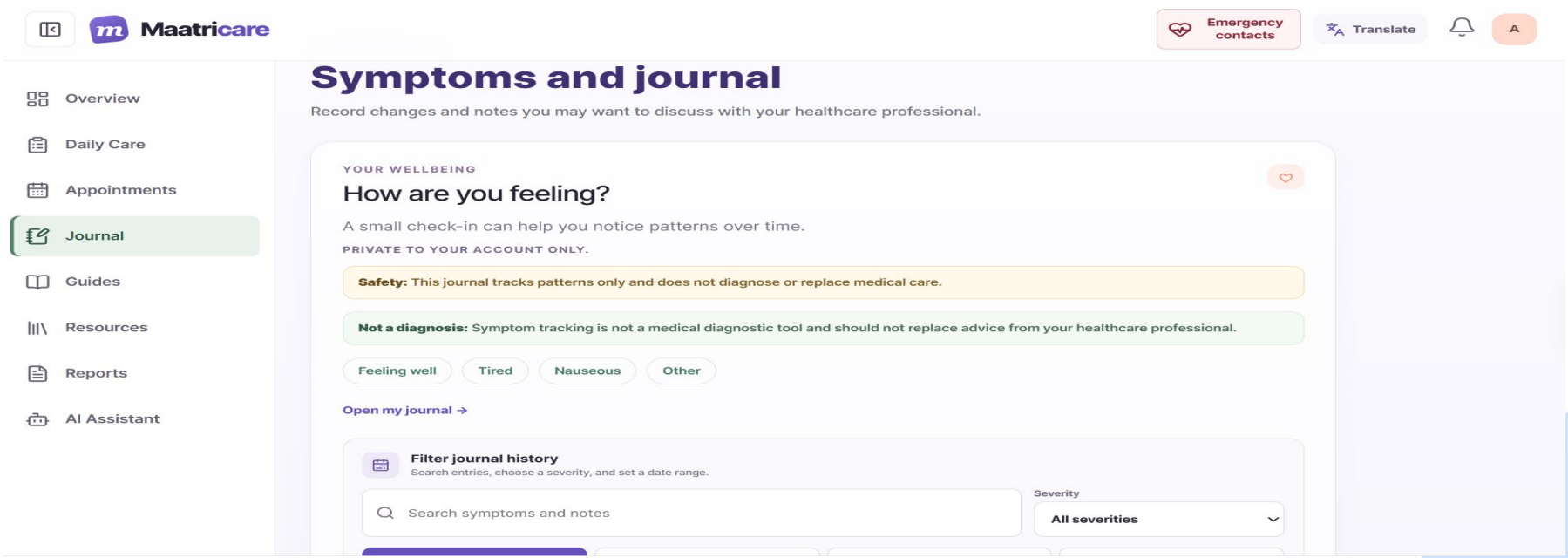
Users can create, search, and review upcoming or past appointments, record provider details, prepare questions, export calendar events, and generate a printable visit brief. The brief includes only information the user explicitly selects.



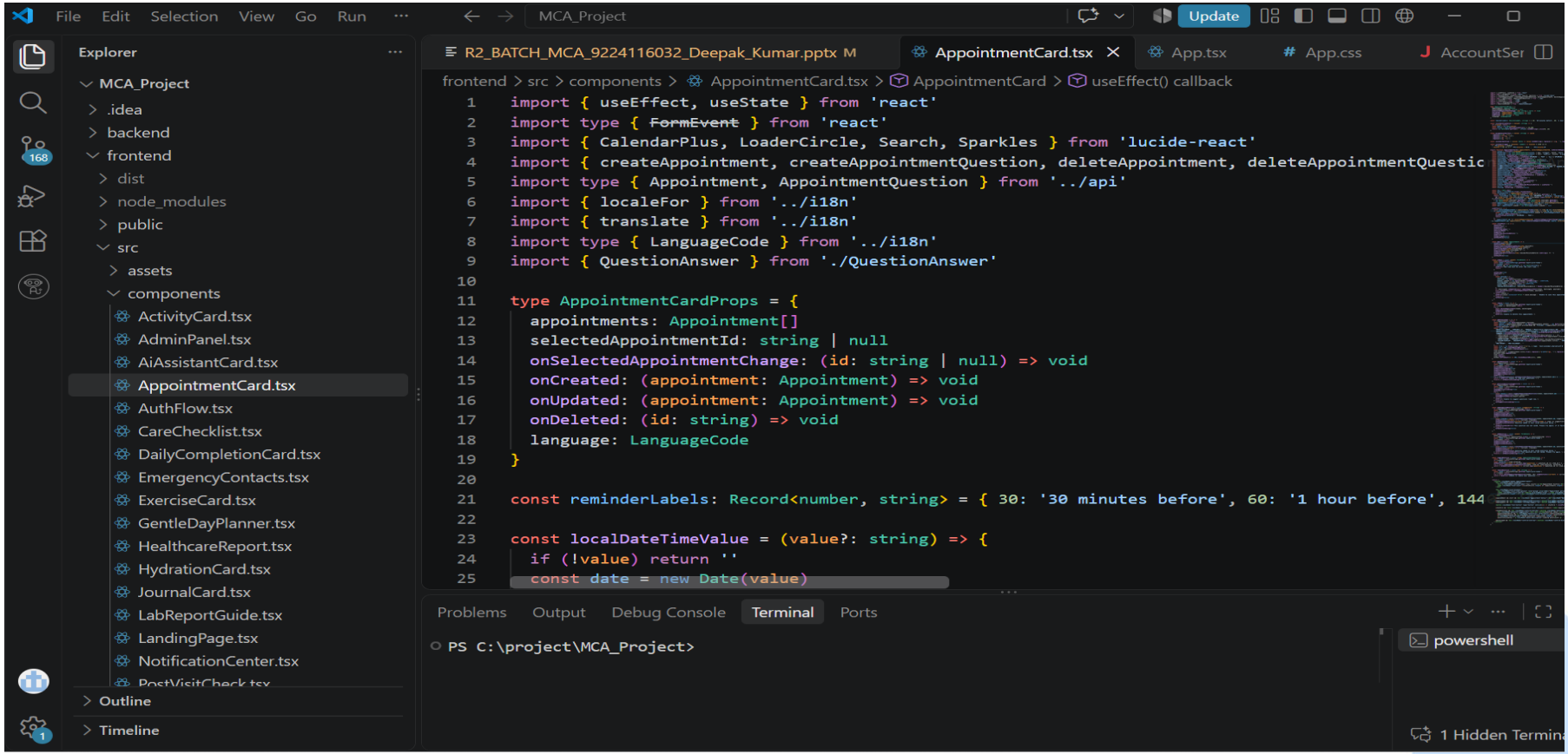
Private Journal and Learning Modules

Journal: record symptoms, severity, dates, and private notes; search and filter your entries.

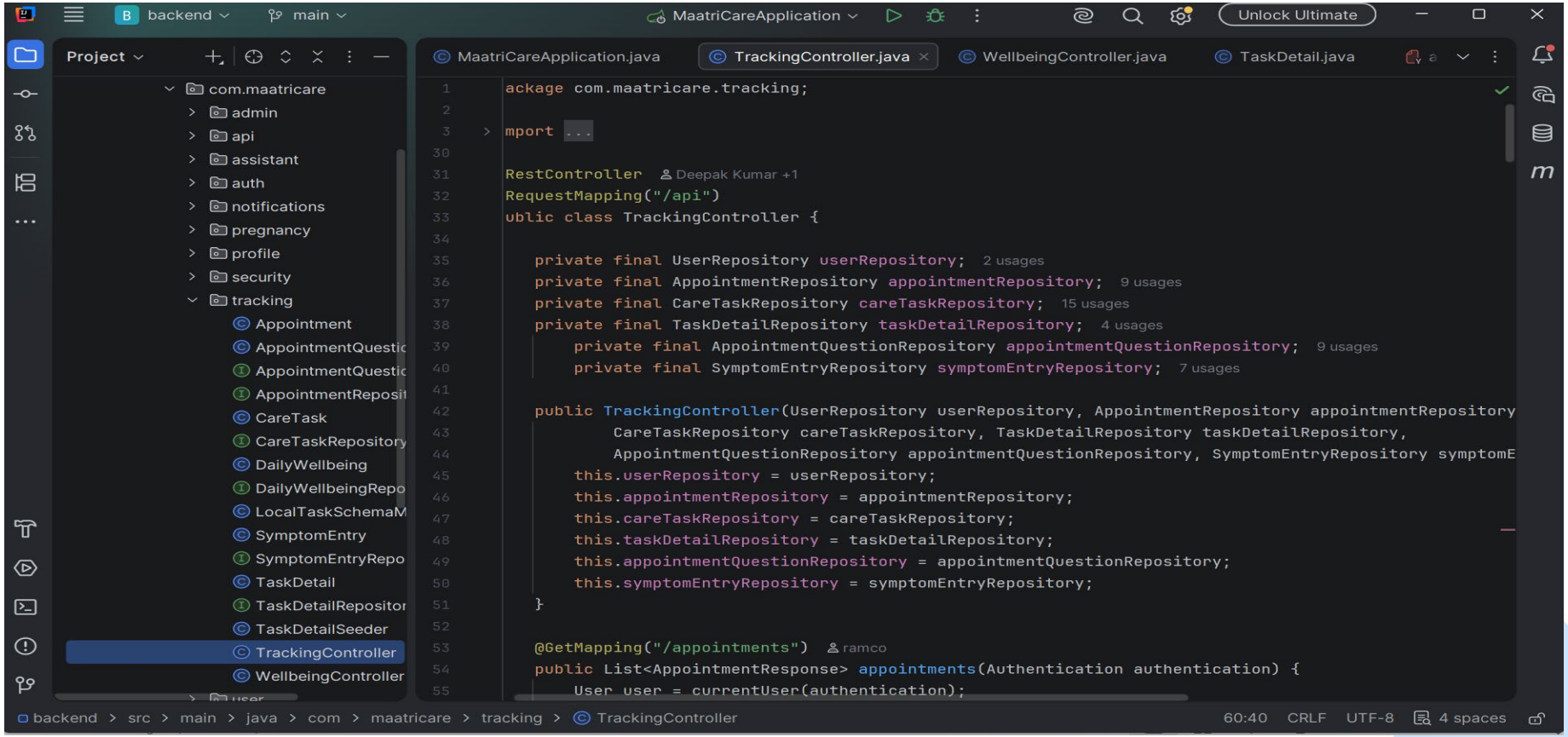
Guides: review general, week-based information and searchable educational resources. Discuss personal questions with a healthcare professional.



Sample Code : UI : TypeScript : AppointmentCard



Sample Code : Backend : Spring Boot : TrackingController



Sample Data Set : MySQL / PostgreSQL

TABLE	SAMPLE FIELDS	PURPOSE
users	id, email, timezone_id	account
appointments	user_id, starts_at, reminder_minutes_before	visit schedule
care_tasks	user_id, task_date, completed	daily tasks
symptom_entries	user_id, symptom, severity	private journal
push_reminders	user_id, key, scheduled_at, status	scheduled delivery
push_subscriptions	user_id, endpoint_hash	opt-in device

All example records are synthetic and contain no real health information.

Testing Methods and Results

- Backend: Spring Boot Test, MockMvc, JUnit, and H2 PostgreSQL mode. Four focused reminder tests passed.
- Coverage: timezone default and validation; schedule preferences; deduplication; one-time expiring snooze; 90-day retention.
- Frontend: TypeScript/Vite production build passed.
- Privacy checks: JWT authentication, ownership checks, selective report sharing, generic push text.
- Limitations: no live push delivery without VAPID keys and HTTPS; broader UI, accessibility, and MySQL integration tests remain future work.

Conclusion and Future Work

MaatriCare provides an authenticated workspace for pregnancy records, appointments, daily care, private journaling, educational resources, user-controlled reports, and reminders. The system focuses on organization and communication support.

Future work

- Configure VAPID and HTTPS, then test push on real devices
- Replace illustrative interface panels with live screenshots
- Extend frontend, database, migration, and accessibility tests
- Complete privacy/security review and production deployment
- Prepare the final report in the official institutional format

Reference Books and Websites

- React Documentation. React.dev.
- Spring Boot Reference Documentation. Spring.io.
- Spring Security Reference Documentation. Spring.io.
- PostgreSQL Documentation. PostgreSQL.org.
- MySQL 8.0 Reference Manual. Oracle.
- MDN Web Docs: Service Worker API and Push API.
- Pressman and Maxim. Software Engineering: A Practitionerâ€™s Approach, 9th ed.
- Sommerville. Software Engineering, 10th ed.
- Verify publication details and retain only sources actually consulted.

Thank You